

Question Paper

Exam Date & Time: 03-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026

Database and Application Security [IT_4060]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyse the interrelationships between the three core areas of operating system security: User Administration, System Vulnerabilities, and E-mail Security. (5)
- A) a) Explain how improper User Administration can amplify the risks posed by the specific Vulnerabilities of OS listed for Windows and UNIX systems.
b) Analyse why E-mail is a particularly effective attack vector for exploiting these system vulnerabilities, linking your answer to the principle of not configuring an email server on a machine with sensitive data.
c) Based on your analysis, what is the overarching security principle that connects these three areas?

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Apply the principles of robust authentication to design a security policy for a new employee. Your policy must include: (3)
- a) One specific rule for password creation that addresses both complexity and length.
b) One example of a Multi-Factor Authentication (MFA) method.
c) A brief explanation of how the MFA method in (b) provides greater security than a password alone.

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Apply the principle of least privilege to database links. Identify two configuration best practices from the provided list that directly enforce this principle and explain how each one mitigates risk. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2) Analyse the synergistic relationship between the "Predefined roles description" and the "Administration staff and management" documents. Explain how, when used together, they create a system of accountability that prevents role confusion and establishes a clear audit trail for user administration. (5)
- A)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) A business needs to retrieve live shipping rates from an external partner's web service directly within their Oracle 9i database. Describe the step-by-step process of how a PL/SQL procedure would use Oracle's built-in packages to accomplish this SOAP callout. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) A new "Customer Relationship Management" software is being introduced in a company. Applying (2)

your knowledge of the components of an Information System, explain which two components must be directly addressed to ensure employees can use this new system effectively and correctly.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 3) Analyse the fundamental challenge in monitoring database link activity by comparing the mechanisms used by Oracle and SQL Server and explain tracking usage of links with diagram . (5)

A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Construct a labelled diagram that illustrates the three-tier structure of a database management system (DBMS), showing the flow of interaction from users to stored data. Clearly label the key components in each tier. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Using the principles outlined in the text, describe one practical step you would take to begin mapping data sources and sinks in a distributed environment with diverse architectures such as mobile databases and log shipping. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Compare and contrast the security implications of using traditional text-based passwords versus graphical passwords. In your analysis, refer to the vulnerabilities associated with brute-force and dictionary attacks, and evaluate which authentication method offers a more secure user experience in a public or shared access environment. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) A company uses a DB2 database and needs to expose its data as a web service. Illustrate the high-level architecture for this by drawing a labelled diagram that includes the following components: WebSphere Application Server (WAS) WORF and SOAP Runtime DADX File DB2 Database. (3)
- Show the logical flow and interaction between these components

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Analyse how the dual role of an operating system as a "Resource Manager" and a "main access point in DBMS" creates specific security challenges that must be addressed by its core functionalities. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 5) Organize the fundamental concepts of application architectures by creating a comparative table. The table must have four rows, one for each major application category. For each category, your table must define its core structural model and specify its primary data interaction purpose. (5)
- A)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) A company is training its employees on cybersecurity. For the Ransom cyber-attack, recommend a specific preventive measure that an employee or the IT department should implement (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Identify and justify a more appropriate defensive strategy (e.g., using sanitization libraries like DOM Purify) that targets the root cause (client-side script execution from unsanitized DOM manipulation). (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	IT_4060:Database and Application Security

#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	13
3	3	CO 3	15
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	IT_4060:Database and Application Security

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	31(62%)
4	Analysis	5	19(38%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)